# Yong-chan Park

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## About Me

I am a Ph.D. student majoring in Computer Science and Engineering at Seoul National University (SNU), advised by Prof. U Kang in Data Mining Laboratory. My research interests include machine learning, tensor analysis, and anomaly detection.

## Research Interests

Machine learning, Tensor analysis, Anomaly detection

## Education

**Ph.D. Student** Mar. 2020 - Present
- Computer Science and Engineering, SNU

**Bachelor of Science** Mar. 2012 - Feb. 2019
- Department of Mathematical Sciences, SNU

## Research Experiences

**A novel approach to Fourier Transform [1]** June 2020 - Present
- Propose Partial Fourier Transform (PFT) that rapidly computes a part of Fourier coefficients of a given time series and demonstrate the accuracy and efficacy of PFT on real-world anomaly detection [1]. PFT achieves 20x faster running time compared to SOTA algorithms.
- Patent pending

**Stock movement prediction [2]** Sep. 2020 - Present
- Propose Data-Axis Transformer with Multi-Level Contexts (DTML) for stock movement prediction that learns the correlations between stocks in an end-to-end way [2]. DTML achieves SOTA accuracy on six datasets collected from US, China, Japan, and UK.

**Software development for patent management** Sep. 2020 - Present
- Implement LSH (Locality Sensitive Hashing) algorithm for finding similar patent data, and a BERT-based model for automatic patent classification.
- Develop a GUI with additional functionality of visualizing results.
- Industry-Academy Cooperation with Daewoo Shipbuilding & Marine Engineering Co., Ltd.

## Publications

[1] **Yong-chan Park**, Jun-Gi Jang, and U Kang. “Fast and Accurate Partial Fourier Transform for Time Series Data.” KDD 2021
[2] Jaemin Yoo, Yejun Soun, **Yong-chan Park**, and U Kang. “Accurate Multivariate Stock Movement Prediction via Data-Axis Transformer with Multi-Level Contexts.” KDD 2021
[3] **Yong-chan Park***, Sangjun Son*, Minyong Cho, and U Kang. “DAO-CP: Data-Adaptive Online CP Decomposition for Tensor Stream.” PLOS One 2022 (*equal contribution)
[4] Jun-Gi Jang, Jeongyoung Lee, **Yong-chan Park**, and U Kang. “Fast and Accurate Dual-Way Streaming PARAFAC2 for Irregular Tensors - Algorithm and Application.” KDD 2023
[5] **Yong-chan Park**, Jongjin Kim, and U Kang. “Fast Multidimensional Partial Fourier Transform with Automatic Hyperparameter Selection.” KDD 2024
[6] Jun-Gi Jang, **Yong-chan Park**, and U Kang. “Fast and Accurate PARAFAC2 Decomposition for Time Range Queries on Irregular Tensors.” CIKM 2024
[7] JinGee Kim, **Yong-chan Park**, Jaemin Hong, and U Kang. “Accurate Stock Movement Prediction via Multi-Scale and Multi-Domain Modeling.” BigData 2024
[8] SeungJoo Lee, **Yong-chan Park**, and U Kang. “Accurate Coupled Tensor Factorization with Knowledge Graph.” BigData 2024
[9] **Yong-chan Park**, Kisoo Kim, and U Kang. “PuzzleTensor: A Method-Agnostic Data Transformation for Compact Tensor Factorization.” KDD 2025

## Under Review

[R1] **Yong-chan Park**, SeungJoo Lee, and U Kang. “Fast and Accurate Online Coupled Matrix-Tensor Factorization via Frequency Regularization.”
[R2] **Yong-chan Park**, SeungJoo Lee, and U Kang. “CoCoNat: Leveraging Cross-Query Context to Enhance Named Entity Recognition.”